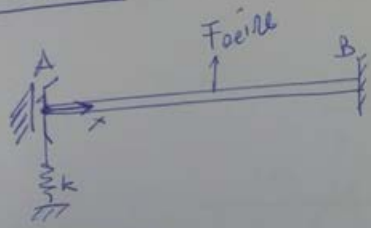


• PROBLEM 2



1) $\vartheta_A = 0 \rightarrow \left. \frac{\partial W}{\partial x} \right|_0 = 0$

2) $T = EI \frac{\partial^3 W}{\partial x^3}$
 $T dx = T_0 + \frac{\partial T}{\partial x} dx$
 $-k W_0 - T_0 - \frac{\partial T}{\partial x} dx = 0$

3) $W(L, t) = 0$

4) $\vartheta_B = 0 \rightarrow \left. \frac{\partial W}{\partial x} \right|_L = 0$

$k W(0, t) + EI \left. \frac{\partial^3 W}{\partial x^3} \right|_0 = 0$

$$\begin{cases} \cancel{x^B} + \cancel{x^D} = 0 \\ kA + kC - EI \gamma^3 B + EI \gamma^3 D = 0 \\ A \cos \gamma L + B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L = 0 \\ -A \cancel{\gamma \sin \gamma L} + B \cancel{\gamma \cos \gamma L} + C \cancel{\gamma \sinh \gamma L} + D \cancel{\gamma \cosh \gamma L} = 0 \end{cases}$$

$H(x) = \begin{bmatrix} 0 & 1 & 0 & 1 \\ k & -\gamma^3 EI & k & \gamma^3 EI \\ \cos \gamma L & \sin \gamma L & \cosh \gamma L & \sinh \gamma L \\ -\sin \gamma L & \cos \gamma L & \sinh \gamma L & \cosh \gamma L \end{bmatrix}$

$V = V_{BEAM} + V_{SPRING} = V_{BEAM} + \frac{1}{2} k W(0, t)^2 \rightarrow K_{m_i} = K_{m-beam_i} + k \phi_i^2(0) \quad i=1, N$

FRF: input and output in $\frac{L}{2} \rightarrow W(x, t) = \underline{\Phi}^T(x) \underline{q}(t)$
 $\rightarrow Q_{m_i} = F_{oeist} \phi_i(\frac{L}{2})$
 $W(\frac{L}{2}, t) = \sum_{i=1}^N \phi_i(\frac{L}{2}) q_i(t)$